

Design and Implementation of a Zone-Based BLE Indoor Localization System Using ESP32

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Abstract—IPS are highly essential in smart infrastructure, automation in industries, and context-aware services where the GPS signal is inaccessible. Bluetooth Low Energy (BLE) has low-power proximity detection capabilities, but has a problem of unstable RSSI due to multipath fading and environmental interference. This paper provides a full edge-based zone localization system using an ESP32-S3 platform. The RSSI measurements are stabilized by using an 8-sample sliding window filter and then thresholds based on zone classification. The general classification accuracy of the experimental validation with 90 RSSI samples in a 9.29 m² indoor setting was shown to have a 91% accuracy. The suggested system will eliminate cloud dependency and show a computationally efficient and scaled indoor proximity solution.

Index Terms—Indoor Localization, BLE, ESP32, RSSI Filtering, Edge Computing, IoT

I. INTRODUCTION

Context-aware IoT services, asset tracking, and smart buildings have become vital topics of research due to the significance of indoor localization. Recent research has shown that there are low-cost BLE-based indoor localization systems that are used in smart infrastructure applications [1]. BLE beacons have also been used to improve signal-based proximity estimation and, more importantly, improve RSSI-based positioning by beacons and the beaconing protocols implemented for the BLE standard [2], [3]

A detailed literature survey of indoor localization technologies identifies the trade-offs between coordinate-trilateration and proximity-classification models of location inherent in interior environments, as well as between localization and mapping models, respectively [4]. An extensive literature review of indoor localization technologies emphasizes the trade-offs between coordinate-trilateration and proximity-classification models of location inherent in interior environments between localization and mapping models, respectively. Edge-oriented IoT architectures have also been suggested to decrease latency and remove cloud reliance on nearby location structures [5].

Regardless of such breakthroughs, it is true that RSSI measurements are very vulnerable to multipath fading, environmental interference, and temporary fluctuations of signals. Thus, signal stabilization methods have been found to enhance the reliability of signal classification of the signal through the methods of signal stabilization used to improve the reliability of the signal classification under investigation [6]. However, most existing systems need multiple anchors, calibration overhead, or computationally intensive filtering models.

This paper will offer a lightweight zone-based BLE indoor localization system to overcome limitations and be built entirely on an ESP32-S3 platform. The proposed system is capable of providing strong proximity detection with low computational complexity and energy consumption by combining RSSI filtering by sliding window and threshold-based zone classification.

The main contributions include the following.

- Complete BLE scanning and RSSI processing on ESP32-S3.
- RSSI stabilization based on Sliding windows .
- Threshold-driven three-zone classification model.
- Experimental validation with 91% classification accuracy.

II. SYSTEM ARCHITECTURE

The proposed system will be based on a BLE beacon that will act as an advertising device and an ESP32-S3 receiver that will handle the scanning and signal processing process. The system workflow consists of the RSSI acquisition process, filtering, and zone classification.

Everything is done locally, which avoids overheads in cloud communications, and enhances scalability in standalone deployments.

A. RSSI Stabilization Model

In order to alleviate sharp instantaneous signal variations, an 8-sample sliding window average is used:

$$\bar{P}(k) = \frac{1}{N} \sum_{i=0}^{N-1} P(k-i), \quad N = 8 \quad (1)$$

This averaging reduces short-term noise and causes classification to be more stable.

III. ZONE CLASSIFICATION FRAMEWORK

Filtered RSSI values are mapped to proximity zones as follows:

- Zone A (Near): $\text{RSSI} \geq -40$ dBm (<0.5 m)
- Zone B (Mid): -55 dBm $\leq \text{RSSI} < -40$ dBm (0.5–1.5 m)
- Zone C (Far): $\text{RSSI} < -55$ dBm (1.5–3.25 m)

The zone-based technique results in fewer algorithms overhead and allows fast classification to be achieved than coordinate estimation and is well suited to real time embedded system needs.

IV. EXPERIMENTAL EVALUATION

The experiments were carried out on a 100 sq.ft (9.29 m²) indoor test setup.

- Maximum tested distance: 3.25 m
- Total RSSI samples: 90 (30 per zone)
- BLE scan duration: 2 seconds per cycle
- Sliding window size: 8 samples

TABLE I
ZONE CLASSIFICATION PERFORMANCE

Zone	Samples	Accuracy
Near	30	93%
Mid	30	91%
Far	30	89%
Overall	90	91%

Transient misclassifications are greatly diminished by the sliding window filter than in raw RSSI measurements. The most significant gains were observed in the mid-zone, where the signal variance is generally more significant.

V. DISCUSSION

It is shown in the system that lightweight filtering and threshold classification can be used to have reliable proximity information without the need to invoke computationally complex algorithms like Kalman filtering and trilateration.

The built-in architecture guarantees low latency, energy-saving and the applicability of the embedded architecture in smart building automation, access control, and IoT-enabled monitoring systems. But larger-scale environments need to deploy multi-beacon and adaptive filtering methods in order to scale.

VI. CONCLUSION

An indoor localization system based on BLE and zone, which is implemented in ESP32, has been introduced. The system is 91% accurate in a controlled indoor environment by combining the RSSI filter sliding window and threshold-based classification. The current framework for proximity-aware applications is the edge-based framework, which is low-complexity and scalable.

The future improvements consist of multi-anchor extension, active filtering, and integration with power-generating IoT instances.

A. Computational Complexity

The operations suggested in the proposed filter and classification do not need more than the simplest arithmetic calculations, which means that the memory consumption is minimal and the computations do not take a lot of time. The implementation of ESP32 does not need any hardware acceleration to run within the constraints of real-time.

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